

Syllabus

Numerical Methods for Mathematical Finance

Academic Year 2025/2026

General Information

- **Instructor:** Matteo Garbelli
- **Email:** `matteo.garbelli@univr.it`
- **Department:** Department of Computer Science
- **Language:** English - **Credits:** 6 CFU

Objectives

By the end of the course, students will be able to: model financial prices using stochastic processes (BM, GBM, SDEs); implement numerical schemes for stochastic differential equations in Python; apply Monte Carlo and other simulation methods for option pricing; tackle topics in energy finance and get an introduction to machine learning.

Outline

Lecture 1 Brownian Motions and Geometric Brownian Motions

- Brownian Motion (BM), Geometric Brownian Motion (GBM);
- Simulation of BMs, GBMs, and correlated GBMs.

Lecture 2 Numerical Methods for SDEs

- EulerMaruyama and Milstein schemes;
- Simulation of OU, CIR and Heston models.

Lecture 3 Pricing

- Real-world vs risk-neutral measure;
- BlackScholes equation and FeynmanKac theorem;
- Monte Carlo simulation for option pricing.

Lecture 4 Advanced Topics and Machine Learning

- Energy markets;
- Introduction to machine learning in finance;
- Reinforcement learning applications.

Schedule

Classes will take place on **Thursdays**, from **10:30 to 12:30**, in **Room I**:

- 08/01/2026
- 15/01/2026
- 22/01/2026
- 29/01/2026

Office Hours

On appointment (contact via email).

Exam and Evaluation

The exam consists of a project with three components:

1. A written report in PDF (self-contained, including derivations and results);
2. Documented and functional Python code;
3. A Beamer presentation for a 30-minute seminar.

The project must be presented in a seminar to pass the course.

References

- C.W. Oosterlee, L.A. Grzelak, *Mathematical Modeling and Computation in Finance*, World Scientific, 2019.
- M.F. Dixon, I. Halperin, P. Bilokon, *Machine Learning in Finance: From Theory to Practice*, Springer, 2020.
Link: <https://link.springer.com/book/10.1007/978-3-030-41068-1>